Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **Python** and **Flask** microservices, **React**, TypeScript, and Next.js frontends, and machine learning and AI systems on AWS and Kubernetes, with SQL and NoSQL schema design, testing, and monitoring.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized relational **schemas** with row-level security and indexing for a 500+ user platform, tuning queries to cut dashboard latency from 1.2s to 350ms in production for every tenant.
- Restructured fault-tolerant **AI job** orchestration with Python FastAPI workers and failure isolation, processing 1K+ jobs at 3x throughput, and shipped validated integrations through CI/CD at 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built and owned a full-stack product end to end with a **React** frontend and modular **Flask** REST backend on AWS, serving 150+ instructors and 10K+ daily requests.
- Integrated multimodal **AI** models GPT-4o, Claude, and Qwen-VL with OCR fallbacks and triaged production failures from logs, reaching 90%+ grading consistency and cutting instructor feedback turnaround 60% in production.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a GPT-based Decision Transformer in PyTorch for **machine learning**-driven vehicle motion planning, with pipelines over 100K+ trajectory logs, improving long-horizon trajectory planning 18% in closed-loop simulation runs.
- Designed a multi-head attention encoder fusing road maps with vehicle trajectories, validating it against three baselines in **closed-loop simulation** and improving closed-loop planning stability 15% while reducing off-road trajectories.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built **object-oriented** Java and Python microservices with Singleton and Factory patterns, validated by 20+ unit and integration **test** suites in nightly Jenkins pipelines across Windows, Linux, and macOS.
- Improved **API performance** with pagination, field filtering, and caching, cutting response times 30%, and redesigned AWS Glue and S3 data pipelines for 68% higher efficiency and 15% lower storage costs.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **Flask** APIs for a two-sided booking marketplace with Firebase **NoSQL** real-time sync, keeping responses under 200ms and preventing double-booked slots with transactional locking across 50K+ records.
- Built a **React and TypeScript** dashboard and payment webhooks for refunds and disputes, cutting manual handling about 70%, and presented the end-to-end architecture to client stakeholders before implementation.

Skills

Languages: Python, Java, TypeScript, JavaScript, C++, SQL, HTML, CSS
Backend: Flask, FastAPI, Spring Boot, REST APIs, Microservices, Object-Oriented Design
Frontend: React, Next.js, TailwindCSS, React Native, Git, Code Reviews, Debugging, Monitoring, Agile
Data & Cloud: PostgreSQL, MySQL, MongoDB, Firebase, NoSQL, Schema Design, AWS, Kubernetes, Docker
ML & Practices: PyTorch, Machine Learning, LLM Systems, Data Structures, Algorithms, Unit Testing, PyTest, CI/CD

Projects

Ride Dispatch and ETA Service | *Python, Flask, DynamoDB, SQS, Kubernetes, PyTest*

- Designed a **Flask** microservice that matches ride requests to nearby drivers, stores trip state in **DynamoDB**, queues status events through SQS, and predicts ETAs with a gradient-boosted model.

Rider Trip Tracker | *Next.js, React, TypeScript, Flask, Jest*

- Designed a **Next.js** and TypeScript web app showing live driver location, trip status, and fare breakdowns from a Flask API, with Jest tests and documented **component** architecture.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, WebSockets*

- Built a **Next.js** and TypeScript platform deployed on **Kubernetes** with CI/CD, Plaid bank linking, and real-time updates, supporting 50,000+ users and 1M+ transactions with MongoDB and transaction filtering.